



Darshan UNIVERSITY

Python Programming - 2301CS404

Lab - 5

Roll No.:315

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELLO123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [5]: s1 = input("enter any string")
countl = 0
countu = 0
countd = 0
counts = 0
for i in s1 :
    if i.islower():
        countl += 1
    elif i.isupper():
        countu += 1
    elif i.isdigit():
        countd += 1
    else:
        counts += 1
print(f"lower is {countl} , upper is {countu} , digit is {countd} , spacial is {counts}")
```

lower is 11 , upper is 2 , digit is 6 , spacial is 3

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE
- Input: Hello World → olleH dlroW
- Input: Learn Python Fast → nrael nohtyP tsaF

```
In [9]: s1 = input("enter any string")
        print(s1[::-1])
```

inamihB avhsiv

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [11]: s1 = input("enter any string")
        if s1.islower():
            print("lower case")
        elif s1.isupper():
            print("Upper")
        else:
            print("Title Case")
```

Title Case

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [13]: s1 = input("enter any string")
        search = input("enter your accours char:")
        fo = s1.find(search , 0)
        lo = s1.find(search , -1)
        print(f"first occurnce is {fo} , last occurence is {lo}")
```

first occurnce is 2 , last occurence is 6

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [18]: s1 = input("enter any string")
search = input("enter your accours char or string:")
s1 = s1.lower();
search = search.lower();
count = s1.count(search)
print(f"frequency counter is {count}")
```

frequency counter is 2

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [26]: s1 = input("enter any string")
s2 = s1.split()
s3 = " ".join(s2)
print(s3)
```

Python is Fun 111

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [27]: s1 = input("enter any string")
pre = input("enter your accours prefix:")
post = input("enter your accours sufux:")
s1 = s1.removeprefix(pre)
s1 = s1.removesuffix(post)
print(s1)
```

happy

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [35]: s1 = input("enter your string :")
s1=s1.lower()
for i in s1:
    if( i == 'a' or i == 'e' or i == 'i' or i == 'o' or i == 'u'):
        s1 = s1.replace(i , "*")

print(s1)
```

v*shv*

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

```
In [21]: s1 = input("enter your string :")
output = ""
i = 0
while i < len(s1):
    c = s1[i]
    count = 0
    while i < len(s1) and s1[i] == c :
        count += 1
        i += 1
    output += c + str(count)

print(output)
```

a2b2c3

10) Toggle Case (without using str.swapcase())

- PyThOn → pYtHoN
- HEllO → heLL0
- Python → pYTHON

```
In [42]: s1 = input("enter your string :")
s2 = ""
for i in s1:
    if i.islower():
        s2 += i.upper()
    elif i.isupper():
        s2 += i.lower()
    else:
        s2 += i
print(s2)
```

vISHvA

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid

- 123User → Invalid
- Us@12 → Invalid

```
In [60]: s1 = input("enter your string :")
s2 = ''
if(s1[0].isdigit()):
    print("invalid")
elif(len(s1)>5 and s1.isalnum() ):
    print("valid")
else :
    print("invalid")
```

invalid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

```
In [64]: s1 = input("enter string")
s1 = s1.lower()
s2 = s1[::-1]
if s1==s2:
    print("Palindrom")
else:
    print("not Palindrom")
```

not Palindrom

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [10]: s1 = input("enter string")
word = max(s1.split() , key=len)
print(word)
```

intresting

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING
 - Lower: python programming
 - Title: Python Programming
 - Capitalize: Python programming

```
In [50]: s1 = input("enter string")
print(s1.lower())
print(s1.upper())
print(s1.title())
print(s1.capitalize())
```

```
vishva bhimani
VISHVA BHIMANI
Vishva Bhimani
Vishva bhimani
```

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

```
In [6]: s = "apple,banana,grape"
s2 = "10,20,30"
s3 = "A,B,C,D"
li = s.split(",")
li2 = s2.split(",")
li3 = s3.split(",")
print(",".join(li))
print("|".join(li2))
print("-".join(li3))
```

```
apple,banana,grape
10|20|30
A-B-C-D
```

```
In [ ]:
```